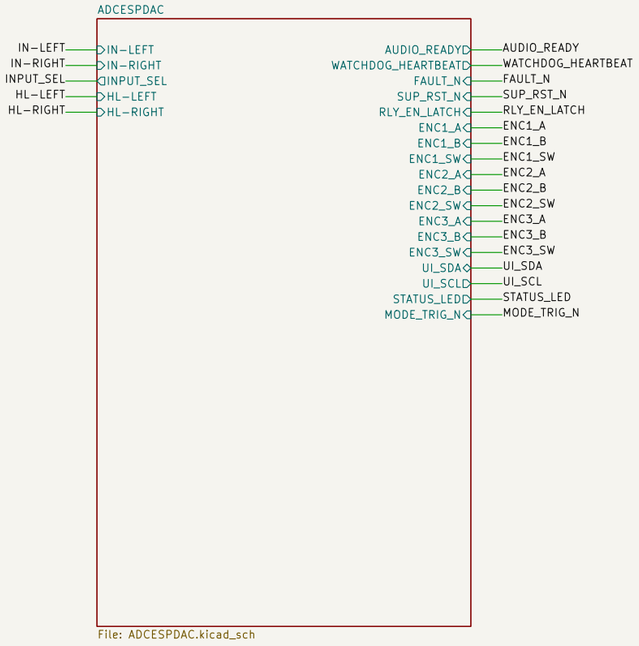
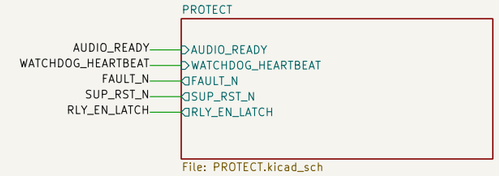
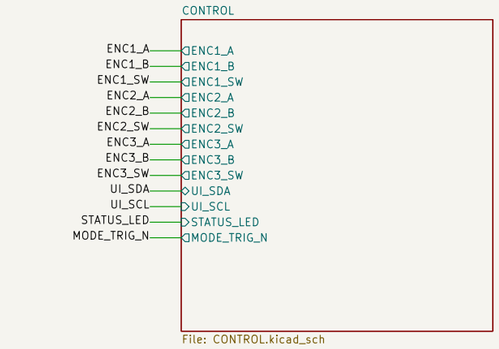








DESIGN NOTES (full engineering report: SubProcessorDigital/DESIGN_REVIEW.md)

Signal: preamp L/R or Marantz LFE -> K2/K3 relays (signal AND source-ground switched) -> 4k7 -> 10u NP -> 100R/10n -> PCM1863.

Clocks: ESP32-S3 I2S0 master, full duplex: BCK 3.072 MHz, LRCK 48 kHz. PCM1863 and PCM5102A both use internal BCK-PLLs.

Latency (fixed, excl. filter group delay): ADC low-latency IIR 10/fs + DMA in 1 block + processing 1 block + DMA out
1-2 blocks + DAC low-latency 3.5/fs. With 32-frame blocks at 48 kHz - 2.3-3.0 ms; 16-frame - 1.3-1.6 ms. MEASURE.

Protection: K1 energises only with latch Q (cleared by any supervisor/watchdog/power fault) AND AUDIO_READY level.

Power: +15 V -> 1N5819 -> LMR51430 (500 kHz FPWM) -> +5 V -> TPS7A2033 x4 (DAC, ADC-A, ADC-D, UI) + TS1117B (ESP32).

GPIO: IO1 XSMT, IO2 WDI, IO3 RLY_SENSE, IO4/5/6 ENC1 A/B/SW, IO7 I2S DIN, IO8 ENC2 SW, IO9 MCLK(opt), IO10 BCK, IO11 LRCK,
IO12/13 I2C0 SDA/SCL (ADC), IO14/21 I2C1 SDA/SCL (UI), IO15/18 ENC2 A/B, IO16 AUDIO_READY, IO17 VBUS sense, IO19/20 USB,
IO38 I2S DOUT, IO39/40/41 ENC3 A/B/SW, IO42 FAULT_N in, IO43/44 UART0, IO45 INPUT_SEL, IO46 STATUS_LED, IO47 MODE_TRIG_N,
IO48 ADC_INT. IO35/36/37 unused (PSRAM on N8R8). IO0 BOOT. Wi-Fi/BT unused (U.FL left open).

Schematic completed/reviewed 2026-10-14: not yet bench tested; PCB not laid out
PCM1863 ADC -> ESP32-S3 IIR DSP -> PCM5102A DAC -> protected output relay -> ESP P68

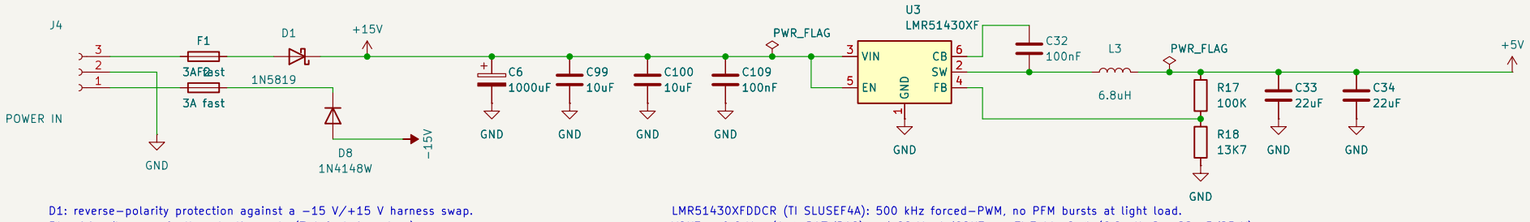
SubProcessorDigital

Sheet: /
File: SubProcessorDigital.kicad_sch

Title: P68 Subwoofer DSP Controller

Size: A3	Date: 2026-10-14	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 1/6

+15 V INPUT, PROTECTION AND 5 V BUCK

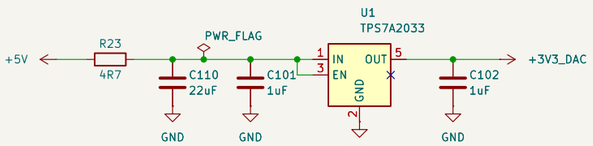


D1: reverse-polarity protection against a −15 V/+15 V harness swap.
F1: wiring/harness fault protection only (3 A fast, inventory).
−15V: HIGH-LEVEL build only (U16). D8 blocks a swapped harness.

LMR51430XFDDCR (TI SLUSEF4A): 500 kHz forced-PWM, no PFM bursts at light load.
VOUT = 0.6 V x (1 + R17/R18) = 4.98 V. L/COUT per TI Table 9-1 (6.8 uH, 2 x 22 uF/25 V).
CIN >= 4.7 uF X7R 50 V + 100 nF at VIN/GND pins (TI 9.2.2.6); C109 closest to U3.
Layout: keep SW node (U3.2-L3-C32) compact; FB divider away from L3; C33/C34 GND to U3 GND.

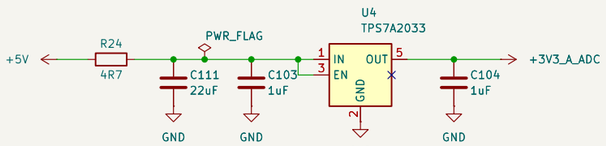
LINEAR REGULATORS (all fed from +5V)

DAC SUPPLY (+3V3_DAC)



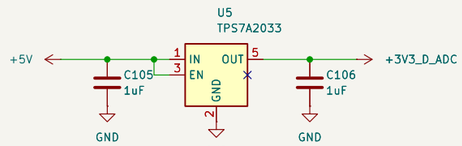
RC pre-filter R23/C110: fc = 1.5 kHz

ADC ANALOGUE SUPPLY (+3V3_A_ADC)

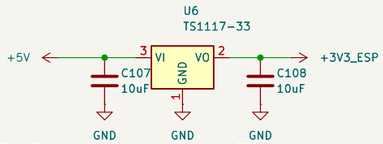


RC pre-filter R24/C111: fc = 1.5 kHz

ADC DIGITAL SUPPLY (+3V3_D_ADC)

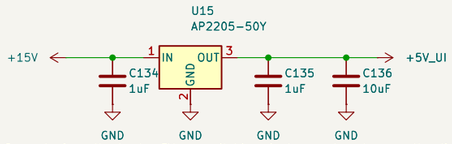


ESP32 SUPPLY (+3V3_ESP)



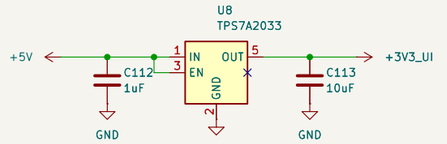
TS1117B: 10 uF MLCC in/out required (TSC datasheet I2405). Pd = 1.7 V x 0.15 A = 0.26 W typ; give SOT-223 tab >= 2 cm2 copper. 22 uF + 100 nF also at module 3V3 pin (ADCESPDAC sheet).

5 V DISPLAY SUPPLY (+5V_UI, J6 pin 20)



Separate from the buck +5V: no switching ripple or relay steps on the display, and a ribbon fault cannot pull down the main rail. Pd = 10 V x I: keep display <= 50 mA (0.5 W, give the tab copper).

UI / DAUGHTERBOARD SUPPLY (+3V3_UI)



Separate LDO: a ribbon-cable fault cannot pull down +3V3_ESP (current-limited 160 mA).

TEST POINTS



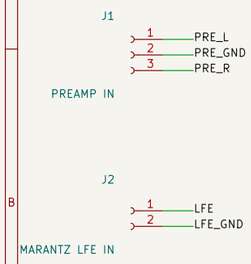
TP8 near U3/C33 (power ground); TP9 near the ADC (quiet ground).

Rails are global power symbols
+15 V in -> LMR51430 5 V buck -> LDOs

P68 Subwoofer DSP Controller

Sheet: /POWER/		
File: POWER.kicad_sch		
Title: Power supply		
Size: A3	Date: 2026-10-14	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 2/6

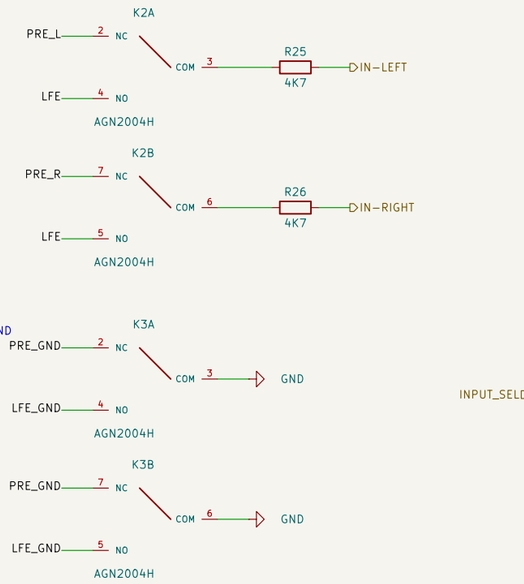
SOURCE CONNECTORS



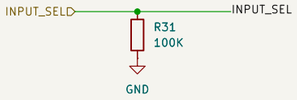
PRE_GND and LFE_GND are NOT connected to board GND
except through the selected K3 contacts.

SOURCE SELECT RELAYS

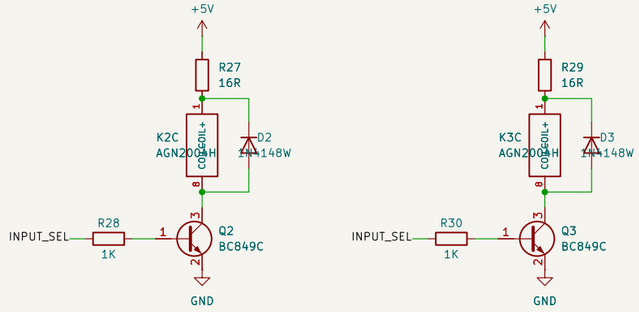
Contacts shown de-energised = PREAMP (stereo) selected.



K2: signal poles. K3: source-ground poles (both poles paralleled).
Energised (INPUT_SEL high) = MARANTZ LFE on both ADC channels; the preamp is then fully disconnected.
Bleed resistors (100k) are on the ADC side of R25/R26 (ADCESPDAC sheet), referenced to board GND,
so no resistor bridges an unselected source to board GND.
R25/R26 4k7: input attenuation/protection. OR = 2.1 Vrms FS, 4k7 = 3.1 Vrms, 10k = 4.2 Vrms.



RELAY DRIVERS



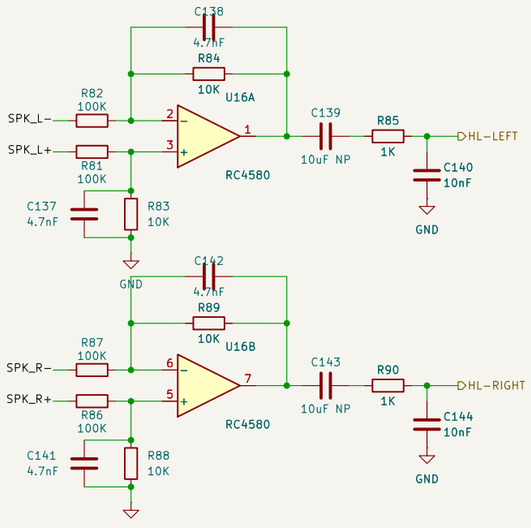
Coil: 145 ohm +/-10 %, 31 mA. 5 V - Vce(sat) via 16R gives ~4.3 V (96 % nom.); must-operate <= 75 % (3.38 V).
Polarised GN coil: pin 1 (+) to supply. 1N4148W clamps turn-off spike (release <= ~4 ms + diode delay).
R31 holds INPUT_SEL low during ESP32 reset (GPIO45 strap must be low at boot).

HIGH-LEVEL (SPEAKER) INPUT

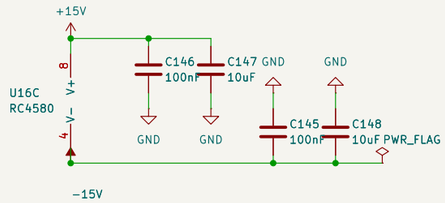
Assembly option: fit EITHER this block (HIGH-LEVEL build) OR J1/J2/K2/K3 + drivers (LOW-LEVEL build). See Config field.
Feeds PCM1863 VINL3/VINR3; firmware selects ADC input VIN3 and sums L+R to mono. No relays.



Speaker terminals: neither side is ground.
Fitted for <= 50 W: 20 Vrms (8R) / 14 Vrms (4R).



Difference amplifier on +/-15 V: Vout = (Rf/Rin)(SPK+ - SPK-), referenced to GND. Zin = 200k differential, 110k each leg to GND.
Common mode: matched 100K/10K pairs; worst-case CMRR ~35 dB with 0.5 % 100K (0.1 % 100K -> ~49 dB). Cf/Cg remove Class-D residue.
Higher-power amps: change Rf = Rg (and Cf = Cg) together: 6K8/6.8nF for 100 W, 4K7/10nF for 200 W into 8R. Rin stays 100K.



C146 at U16 pin 8, C145 at U16 pin 4.

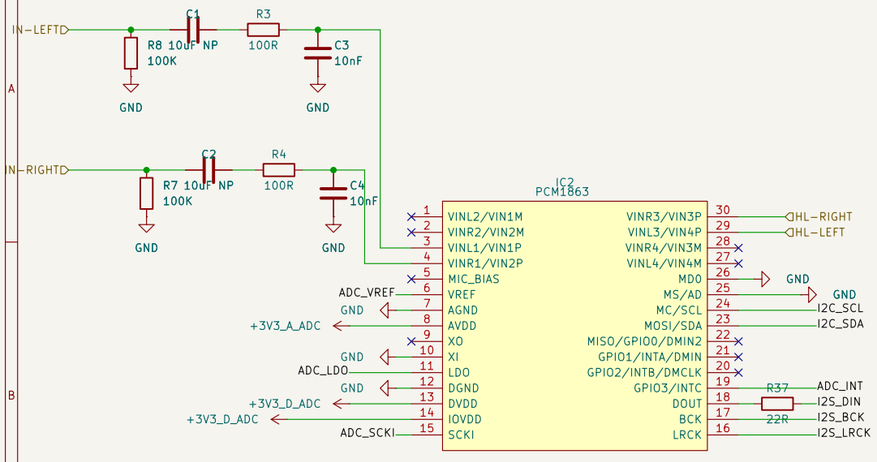
De-energised = PREAMP (stereo)
Relay-switched source signals AND source grounds
P68 Subwoofer DSP Controller

Sheet: /AUDIO-IN/
File: AUDIO-IN.kicad_sch

Title: Audio input selection

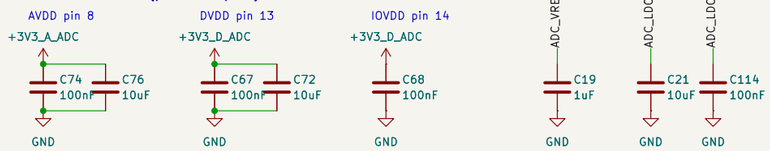
Size: A3	Date: 2026-10-14	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 3/6

ADC PCM1863 (I2C 0x4A, I2S slave, BCK-PLL)

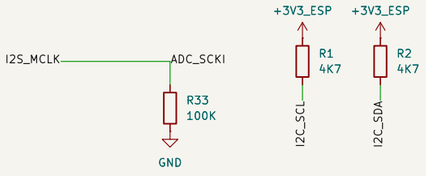


XI tied to DGND: XI and SCKI clocks are OR-ed internally (SLAS831D 9.3.9).
SCKI held low by R33 = BCK-PLL slave mode (Table 15). Fit R32 to use ESP32 MCLK instead.
MD0 = 0: I2C control. MS/AD = 0: address 0x4A. Low-latency IIR decimation = 10 samples.
Input FS 2.1 Vrms single-ended per pin; Zin 10k; TI filter: 10 uF + 100R + 10 nF film/C0G at pin.

ADC DECOUPLING (place at pins)



ADC CLOCK OPTION / I2C0 PULL-UPS



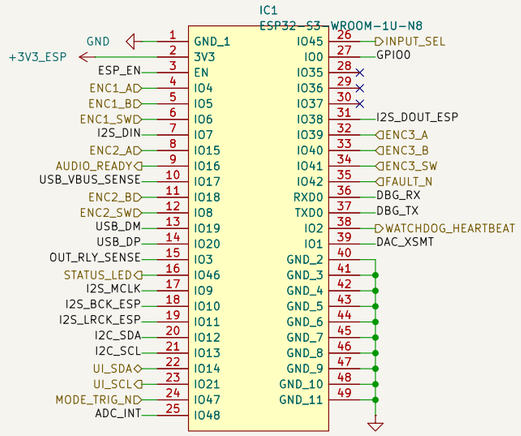
DEBUG UART0 (115200 8N1, 3.3 V)



J8-1 = ESP32 TXD0 (out), J8-2 = ESP32 RXD0 (in)

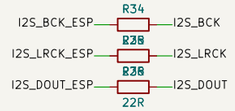
ESP32-S3-WROOM-1U-N8

No PSRAM (85 degC rating). IO35-37 left free, so the N8R8 also fits.



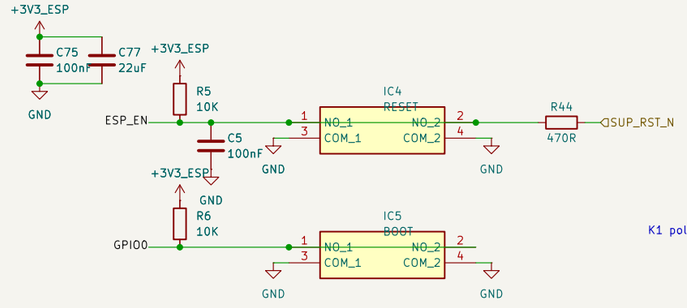
GPIO map: root sheet notes. IO45/IO46/IO0 are strapping pins (low/high-Z at reset).
I2S0 full duplex, ESP32 master: BCK 3.072 MHz (64fs), LRCK 48 kHz. MCLK (IO9) optional.

I2S SOURCE TERMINATION (place R34-R36 within 5 mm of ESP32 pins)

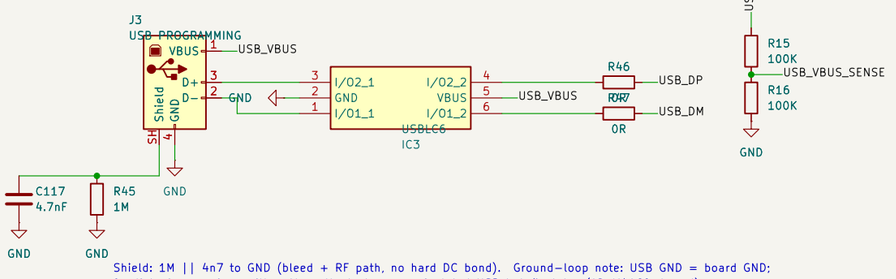


ESP32 SUPPLY / RESET / BOOT

3V3 pin 2: 22u + 100n (Espressif)



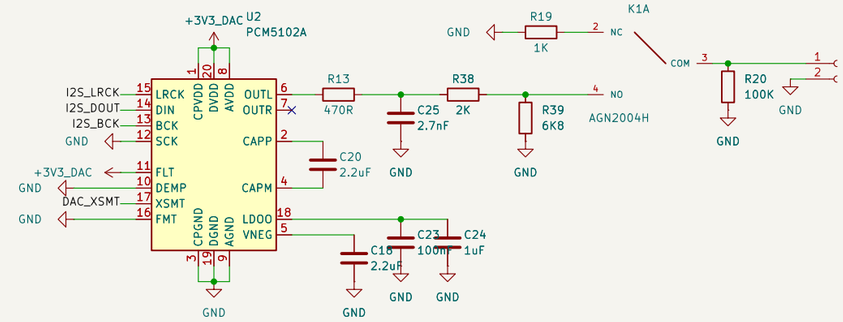
USB (native ESP32-S3 USB-Serial/JTAG; self-powered, VBUS only sensed)



Shield: 1M || 4n7 to GND (bleed + RF path, no hard DC bond). Ground-loop note: USB GND = board GND;
for listening sessions with a Mac attached, use an isolated USB hub/isolator (ADuM4160-based).

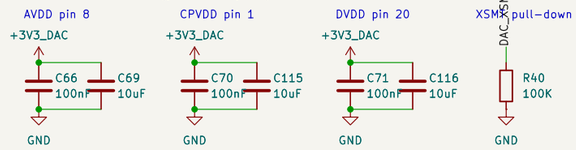
DAC PCM5102A

Hardware mode: I2S, low-latency filter, de-emphasis off, BCK-PLL (SCK = GND).

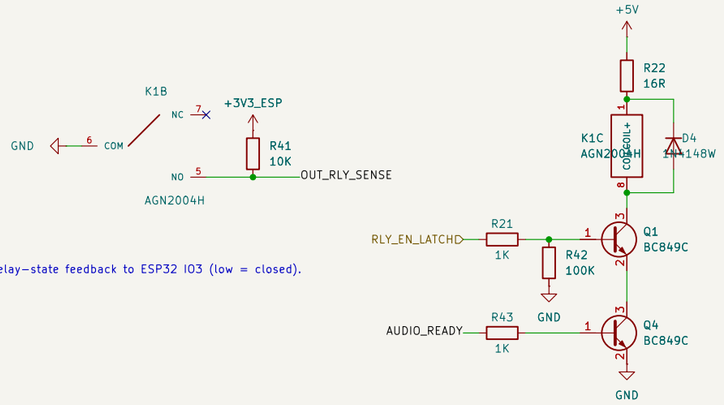


K1 de-energised (no power / fault / not armed): P68 input = GND via R19 1k + R20 100k.
Mute sequence: XSMT low (104-sample soft mute) -> wait 5 ms -> drop AUDIO_READY (relay opens).
Unmute: AUDIO_READY rising edge arms latch -> relay closes (DAC muted, 0 V) -> wait 20 ms -> XSMT high.

DAC DECOUPLING (place at pins)



OUTPUT RELAY DRIVER (two series switches = hardware AND)



K1 pole B = relay-state feedback to ESP32 IO3 (low = closed).

Relay energises only if: latch Q (PROTECT) = 1 AND AUDIO_READY = 1 AND +5V present.
Coil 145R +/-10 %: ~4.3 V across coil via R22 (96 % nom.), must-operate 75 %. D4 clamps turn-off.

Output relay K1 hardware-qualified by latch (PROTECT sheet) AND AUDIO_READY
PCM1863 (I2S slave, BCK-PLL) -> ESP32-S3 -> PCM5102A (BCK-PLL)

P68 Subwoofer DSP Controller

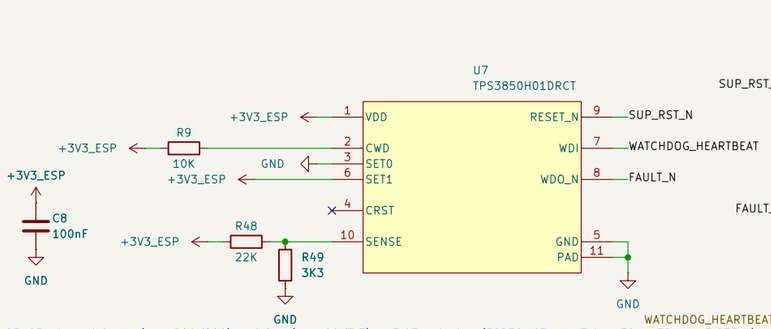
Sheet: /ADCESPDAC/
File: ADCESPDAC.kicad_sch

Title: ADC / ESP32-S3 DSP / DAC / output relay

Size: A3	Date: 2026-10-14	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 4/6

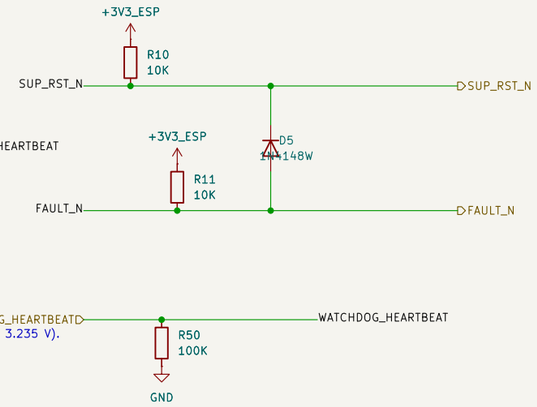
ESP32 SUPERVISOR + WINDOW WATCHDOG

TPS3850H01 (TI SBVS301B), DRC0010J VSON-10

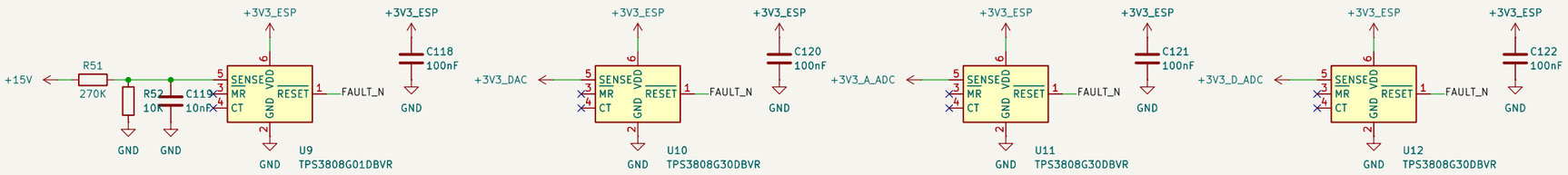


SENSE trip = 0.4 V x (1 + R48/R49) = 0.4 x (1 + 22/3.3) = 3.07 V falling (ESP32-S3 min 3.0 V; TS1117B min 3.235 V).
SET0 = 0, SET1 = 1, CWD = 10k: window 10.35 ms (max lower) .. 165.8 ms (min upper). Kick WDI every ~50 ms
from the audio task only while audio is verified healthy. CRST open: tRST = 200 ms (RESET and WDO pulse).
RESET -> ESP32 EN (via 470R). WDO -> FAULT_N only (not EN): ESP32 boot time exceeds any useful window.

FAULT AND RESET NETS



RAIL SUPERVISORS (TPS3808, SENSE independent of VDD: a dead rail is still reported)



+15V falling < 0.405 x (1 + 270/10) = 11.3 V: early power-fail warning.
C6 (1000 uF) then holds 5 V for ~18 ms at full load: relay is open well before any rail sags.

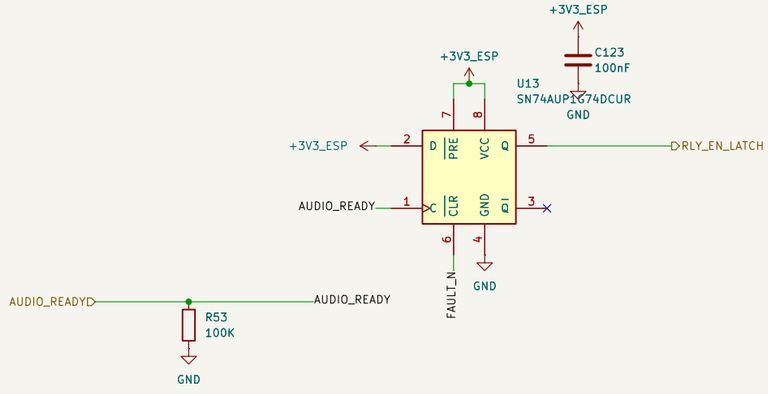
trips if +3V3_DAC < 2.79 V (TPS3808G30)

trips if +3V3_A_ADC < 2.79 V (TPS3808G30)

trips if +3V3_D_ADC < 2.79 V (TPS3808G30)

All RESET outputs are open-drain, wired-OR on FAULT_N (R11 pull-up). CT open = 20 ms release delay.

OUTPUT-RELAY FAULT LATCH (SN74AUP1G74, Schmitt inputs)



LATCH OPERATION

FAULT_N low (any of: TPS3850 RESET via D5, TPS3850 WDO, U9 +15V, U10 DAC, U11 ADC-A, U12 ADC-D) -> Q = 0.
Q = 1 only on a RISING edge of AUDIO_READY while FAULT_N is high (D = 1).
Relay coil current needs BOTH Q (Q1) and AUDIO_READY level (Q4) - see ADCESPDAC sheet.
A GPIO stuck high produces no new edge: after any fault the relay stays open until firmware
drops AUDIO_READY and raises it again (after re-verifying clocks, ADC data and FAULT_N on IO42).
Power loss: +15V monitor trips first; latch, Q1/Q4 and coil all fail to the de-energised state.
Firmware arm sequence: kick WDI >= 3 windows -> read FAULT_N = 1 -> AUDIO_READY 0->1 ->
check OUT_RLY_SENSE (IO3) low within 10 ms -> wait 20 ms -> DAC XSMT high -> ramp volume.

A stuck-high GPIO cannot hold the relay after a watchdog/rail fault
Any fault clears the latch; only a new AUDIO_READY rising edge can re-arm it

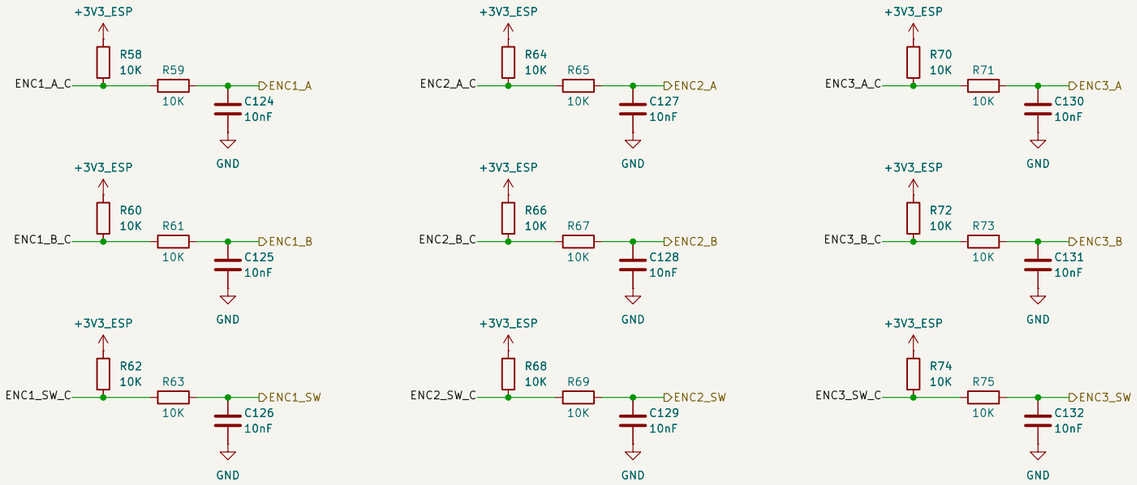
P68 Subwoofer DSP Controller

Sheet: /PROTECT/
File: PROTECT.kicad_sch

Title: Protection: supervisors, watchdog, output-relay fault latch

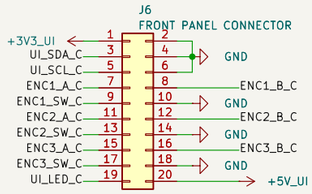
Size: A3	Date: 2026-10-14	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 5/6

ENCODER / SWITCH CONDITIONING (Bourns PEC11R suggested filter: 10k pull-up, 10k series, 10 nF)



Inputs read by ESP32 PCNT (quadrature) + GPIO; firmware debounce on switches. tau = 100 us.

DAUGHTERBOARD HEADER J6 (2x10 shrouded IDC, short ribbon)

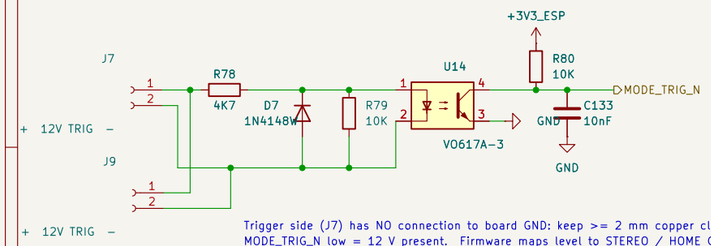


Pinout (ribbon order): 1 +3V3_UI, 2 GND, 3 SDA, 4 GND, 5 SCL, 6 GND, 7 E1A, 8 E1B, 9 E1SW, 10 GND, 11 E2A, 12 E2B, 13 E2SW, 14 GND, 15 E3A, 16 E3B, 17 E3SW, 18 GND, 19 LED, 20 +5V_UI (5 V display). I2C wires each flanked by GND; encoders: PEC11R (switch to GND), common to GND. Display: 3.3 V type from pin 1, or 5 V type (with level shifter, 3.3 V-side pull-ups only) from pin 20.

UI I2C (I2C1, separate from ADC I2C0)

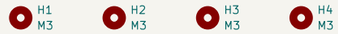
STATUS LED (IO46)

12 V MODE TRIGGER INPUT (optically isolated)



Trigger side (J7) has NO connection to board GND: keep >= 2 mm copper clearance around J7/R78/D7/R79/U14 pins 1-2.
MODE_TRIG_N low = 12 V present. Firmware maps level to STEREO / HOME CINEMA preset and debounces (>= 50 ms).

MOUNTING (M3; isolated — board GND is referenced via the P68 output connector)



12 V trigger is galvanically isolated (TCMT1107)
Controller works with the daughterboard disconnected (all inputs pulled up on this board)

P68 Subwoofer DSP Controller

Sheet: /CONTROL/
File: CONTROL.kicad_sch

Title: Control: daughterboard header, 12 V trigger, LEDs, debug

Size: A3	Date: 2026-10-14	**Rev: A**
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